Luckie Musngi

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EDUCATION

University of Arizona (Graduated May 2026) **Tucson, AZ**

- **Bachelor of Science** in Computer Science, **Minor** in Mathematics
- **Relevant Coursework:** Operating Systems, Database Design, Computer Networks, Web Programming

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C/C++
Backend & Tools: Node.js, Express.js, PostgreSQL, MongoDB, Git
Data & ML: Pandas, NumPy, Scikit-learn, Matplotlib

EXPERIENCE

University of Arizona **Tucson, AZ**
Undergraduate Researcher, Bio Robotics & Scientific Instrumentation Jan 2026 - May 2026

- Designed and evaluated automated robotic systems for experimental workflows in chemistry environments
- Integrated sensors, actuators, and control systems to support repeatable automated procedures
- Performed system testing and reliability evaluation through iterative experimental validation

Lin's Grand Buffet **Tucson, AZ**
Expeditor July 2022 - Present

- Coordinated high-volume workflow between kitchen stations under time constraints
- Communicated across teams to resolve operational bottlenecks during peak service hours
- Maintained efficiency and accuracy in fast-paced, real-time environment

Vail Academy and High School **Tucson, AZ**
Site Tech Coordinator Shadow Sep 2021 - Nov 2021

- Assisted with troubleshooting hardware, software, and network issues
- Supported technical setup for school events and classroom systems
- Gained exposure to IT support workflows and infrastructure maintenance

PROJECTS

Tucson Crime Analytics & Classification System | https://github.com/LuckieMusngi/Crime-in-Tucson-Analysis

- Built end-to-end data processing pipeline for large-scale crime datasets including cleaning, feature engineering, and transformation
- Developed and evaluated classification models using cross-validation and performance metrics (precision, recall, F1-score)
- Engineered structured dataset outputs enabling statistical trend analysis and predictive modeling
- Visualized crime patterns and model performance for interpretability and reporting

Python | Pandas, NumPy, Scikit-learn, Matplotlib

Environmental Health Data ETL + PostgreSQL System | Air vs. Health RDB

- Designed normalized relational schema integrating messy and unrelated environmental and health datasets
- Built ETL pipeline processing 100K+ records with structured cleaning and transformation logic
- Implemented bulk data ingestion into PostgreSQL using optimized insert operations
- Performed relational queries to analyze correlations between environmental and health indicators

Python | PostgreSQL, psycopg2, Pandas

Ski Resort Relational Database System | https://github.com/LuckieMusngi/Ski-Resort-Database

- Designed normalized relational database schema with constraints and entity relationships
- Implemented CRUD operations using JDBC and SQL query integration
- Built command-line interface for database interaction and management
- Applied ER modeling principles to structure relational data system

Java | SQL, JDBC

Bank Website Management System | https://github.com/LuckieMusngi/BankWebsite

- Built full-stack web application supporting user authentication and session-based workflows
- Developed RESTful backend APIs using Express.js
- Integrated MongoDB for persistent data storage and retrieval
- Implemented multi-page user flow and state management

Node.js | Express, MongoDB

High-Speed Arduino Sorting Machine | https://github.com/LuckieMusngi/Arduino-Skittle-Sorter_Quannel-Method

- Designed and implemented high-speed embedded sorting system with real-time control logic
- Integrated sensors and actuators for automated object classification and routing
- Engineered mechanical system using CAD-designed components for optimized flow
- Achieved throughput of ~4.4 items/sec through timing and control loop optimization

Arduino | C/C++, OpenSCAD

Real-Time Procedurally Generated Game (Unity / C#) | https://github.com/LuckieMusngi/Unity-Roguelike

- Built procedural level generation system for replayable gameplay environments
- Implemented AI behavior systems and real-time combat mechanics
- Designed modular architecture using object-oriented Unity components
- Used coroutine-based event sequencing for runtime game flow control

C# | Unity